Project Proposal: Testing Ordinal Representation Linearity in Dense and Sparse Activation Spaces

Group Project Proposal

Research question. Does a graded concept such as *hedging* or *assertiveness* correspond to an approximately one-dimensional affine coordinate in a language model's internal representation space, and is this approximation stronger in the latent space of sparse autoencoders (SAEs) than in the raw dense residual stream? This project focuses on *representation linearity only*: we will not test steering interventions. Instead, we will ask whether trait *intensity* is geometrically organized along a shared direction.

Motivation. Work on activation steering often assumes that a behavioral property can be summarized by a linear direction, but the evidence is uneven. Some papers give positive cases: the Linear Representation Hypothesis formalizes when concept representations can be linear and connects linear representation to probing and interventions [1]; sentiment has been argued to be largely one-dimensional in several models [2]; and truthfulness has been studied via linear geometric structure [3]. However, most of the broader steering literature evaluates *control success*, not whether natural-language examples of a concept at multiple intensities actually lie on an affine axis. Recent reliability work further suggests that steering often depends on local geometric coherence rather than a robust global linear structure [4].

Prior work. Three threads motivate our setup. First, representation-level work studies linear concept directions in dense activations [1–3]. Second, steering papers such as Activation Addition and CAA exploit residual-space directions but usually test downstream control rather than the geometry of graded concepts [5, 6]. Third, the SAE literature argues that dense activations are superposed and that sparse feature dictionaries may recover more interpretable and more axis-aligned latent features [7, 8]. This motivates a natural comparison: perhaps dense residual space contains only a rough local linear approximation, while SAE coordinates provide a cleaner low-dimensional structure for concept intensity. At the same time, benchmark evidence is mixed: AxBench finds that SAE-based methods are often outperformed by simple representation baselines for concept detection and steering [9]. This makes the dense-vs-SAE comparison both nontrivial and empirically important.

Research gap. Prior work has not cleanly tested *ordinal representation linearity*—whether level 0,1,2,3,4 realizations of the same trait lie on a common axis with approximately parallel increments—and has not compared this property systematically between dense residual space and SAE latent space. Existing papers largely study binary contrasts, intervention success, or interpretability of sparse features, leaving open whether SAEs actually improve the geometry of graded semantic variables.

Hypotheses.

- **H1:** For some graded traits, class centroids across intensity levels are approximately rank-1 in activation space.
- **H2:** This rank-1 structure is stronger in SAE latent space than in dense residual space.
- **H3:** Adjacent intensity increments are more parallel in SAE space than in dense space.
- **H4:** Linear predictors trained on one set of templates transfer better across topics and paraphrases in SAE space than in dense space.

Model choice. Our primary model will be **Gemma 2 2B** with **Gemma Scope** SAEs. This is the best lightweight choice for three reasons. First, Gemma 2 is explicitly positioned as a lightweight open model family, and the 2B model is small enough for interpretability experiments on modest hardware. Second, Gemma Scope releases open SAEs for *all layers and sublayers* of Gemma 2 2B, including residual-stream SAEs across all 26 layers, which is unusually comprehensive [8]. Third, the tooling is mature and widely used. Although Gemma Scope 2 now covers Gemma 3 models from 270M upward, we prefer Gemma 2 2B over Gemma 3 270M because the 2B model is still lightweight while being substantially more plausible for subtle graded semantic structure.

Data. We will build a small benchmark of *graded concept ladders*. Each example will instantiate the same semantic scenario at five intensity levels for one trait. We propose one canonical trait (**hedging/confidence**) and one more behavior-like trait (**assertiveness** or **politeness**). For each trait we will create around 40 base scenarios, each rewritten at five intensity levels and with 2–3 paraphrases per level. Topic and lexical variation will be balanced across levels. A small human or LLM-assisted validation stage will verify that the intended level ordering is perceived correctly.

Representations. For dense space, we will extract residual-stream activations from a subset of layers spanning early, middle, and late computation. For sparse space, we will use the corresponding Gemma Scope residual-stream SAEs and encode the same activations into latent feature vectors. To reduce tokenization noise, we will average activations over the target sentence or clause rather than using a single-token readout. We will initially use one SAE width (e.g., 16k) and optionally test one larger width as a robustness check.

Evaluation. We will pre-register four complementary tests of linearity.

- **Rank-1 centroid structure:** compute centroids for each intensity level and measure variance explained by the first principal component.
- **Increment parallelism:** compare cosine similarity among adjacent difference vectors $c_1 - c_0$, $c_2 - c_1$, etc. A truly affine intensity axis should yield high alignment.
- **Linear predictability:** train a linear regressor from representation to intensity on one subset of templates and test on held-out topics/paraphrases; compare against a shallow nonlinear baseline (kernel ridge or small MLP).
- **Transport across contexts:** learn the best intensity direction on one subset of scenarios and test whether projection preserves level ordering on unseen scenarios.

We will use bootstrap confidence intervals and permutation tests for significance, and we will compare dense and SAE spaces at matched layers.

Controls and ablations. The main confound is lexical shortcutting. We will therefore use multiple paraphrases per level and evaluate cross-template transfer. We will also include a random overcomplete baseline or PCA-expanded dense baseline to test whether any SAE advantage simply comes from higher dimensionality. If time permits, we will compare one easy trait (hedging) and one harder trait (assertiveness/politeness) to see whether SAE advantages depend on trait complexity.

Expected contribution. The project will contribute (1) a benchmark for *ordinal* concept geometry, (2) a direct empirical comparison between dense residual space and SAE latent space, and (3) evidence about whether sparse feature spaces actually provide better low-dimensional affine structure for graded concepts. A positive result would support the idea that SAE spaces are not merely more interpretable but geometrically closer to latent semantic variables. A negative result would be equally informative, suggesting that current assumptions about linear trait axes are overstated even in sparse spaces.

Feasibility. The study is intentionally scoped to one model family, two traits, and a modest custom benchmark. No finetuning is required. The main engineering work is dataset construction and feature extraction, which makes the project realistic for a small course team while still addressing a substantive open question in representation geometry.

References

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